Topics to study and practice (Snippets)

Arithmetic Snippets:

```
int a= 10, b=5, c=0;
c= a++ + ++a - --b;
c= 10 + 12 – 4;
c= 18
```

```
int x=2, y=4, z=5;
z -= --x + ++x + x%5;
z = z- (--x +  ++x  +  y%5)
z = 5- (1 + 2 + 4)
z= 5- 7
z= -2
```

```
int x=4, y=5;
x*=y
x=x*y
x=4*5
x=20
```

Conversion from mathematical expression to JAVA Expression:

$\sqrt{x - bk^{10}}$

Math.sqrt(x- b*Math.pow(k,10));

$\frac{a^x}{|b|}$

Math.pow(a,x) / Math.abs(b)

Evaluation of Mathematical Expression written using Math class' functions

Math.pow(25, 1/2)+ 4

- ⇨ 25^0 +4
- ⇨ 1+4
- ⇨ 5

Math.pow(25, 1.0/2.0) + 4

- ⇨ $25^{0.5}$ + 4
- ⇨ 5 +4
- ⇨ 9

Math.max(3, Math.min(5,4))

- ⇨ Math.max(3, 4)
- ⇨ 4

Writing if-else statement using ternary:

ternary syntax: variable= (condition) ? if true: if false;

```
int b;
if(a>0){
b=1;
}else{
b=2;
}

int b=(a>0)?1:0;
```

```
int b;
if(a>0){
b=1;
}else if(a==0){
b=2;
}else{
b=3;
}

int b= (a>0)? 1: (a==0)?2: 3;
```

Conversion of if-else to switch case

```
if(a==1) System.out.println("Monday");
else if(a==2) System.out.println("Tuesday");
else if(a==3) System.out.println("Wednesday");
else if(a==4) System.out.println("Thursday");
else if(a==5) System.out.println("Friday");
else if(a==6) System.out.println("Saturday");
else if(a==7) System.out.println("Sunday");
else System.out.println("Wrong input");
```

```
switch(a){
case 1: System.our.println("Monday"); break;
case 2: System.our.println("Tuesday"); break;
case 3: System.our.println("Wednesday"); break;
case 4: System.our.println("Thursday"); break;
case 5: System.our.println("Friday"); break;
case 6: System.our.println("Saturday"); break;
case 7: System.our.println("Sunday"); break;
default: System.out.println("Wrong Input");
}
```

Character Snippets

```
char ch= 'B'
int k= ch-1;
char ch2= (char) k;
System.out.println(k);
System.out.println(ch2);

Output: 65
        A
```

```
char ch= 'd';
char ch1= 'e';
System.out.println(ch+ch1);

Output:  201
```

```
char ch1= 'C'
float f= ch1-2.0;
System.out.println(f);

Output: 65.0
```

Conditional Snippets

```
int a=5, b=10;
if(a>5 || b<=10){
System.out.println(true);
}else{
System.out.println(false);
}

Output:  true
```

```
switch(a){
case 1: System.out.println(100);
case 2: System.out.println(200);
        break;
case 3: System.out.println(300);
default: System.out.println(Wrong input);
}
```

for a=1	for a=2	for a=3
100	200	300
200		Wrong Input

***** Absence of break statement causes fall through***

Conversion between for() , while(), do-while() loops

for loop syntax:

```
for(initialization; condition; update-statement){
//code
}
```

```
for(int i=1; i<=10; i++){
System.out.println(i);
}
```

while loop syntax:

```
initialization;
while(condition){
//code
update statement
}
```

```
int i=1;
while(i<=10){
System.out.println(i);
i++;
}
```

do-while syntax:

```
initialization;
do{
//code
update statement;
}while(condition);
```

```
int i=1;
do{
System.out.println(i);
i++;
}while(i<=10);
```

Loop dry run:

```
int p=1;

for(int i=1; i<=4; i++){
p*=i;
}
System.out.println(p);
```

Output: 24

i	p=1
1	p=1*1 =1
2	p=1*2 =2
3	p=2*3 =6
4	p=6*4 =24

```
int i=1, k=2;

while(i++ < 5){
if(i==2)continue;
k*=i;
}
```

i	k=2
2	continue
3	k=2*3 =6
4	k=6*4 =24
5	k=24*5 =96

Sequence:

1 2 5 10 17 26

0^2+1 1^2+1 2^2+1 3^2+1 4^2+1 5^2+1

i^2+1 [0, 5]

```
for(int i=0; i<=5; i++){
System.out.println(i*i + 1);
}
```

0 1 1 2 3 5 8 13 21 …. upto nth term (fibonacci)

```
int n1=0, n2=1;
System.out.print(n1+" "+n2);
for(int i=1; i<=n; i++){
int s=n1+n2;
System.out.print(s+ " ");
n1=n2;
n2=s;
}
```

Series:

S= 1+3+5+7+9+…+ n

```
int s=0;
for(int i=1; i<=n; i+=2){
s+=i;
}
System.out.print(s);
```

factors of a number (n)

```
for(int i=1; i<=n; i++){
	if(n%i==0){
	System.out.println(i);
	}
}
```

Count of factors of a number (n)

```
int c=0;
for(int i=1; i<=n; i++){
	if(n%i==0){
	c++;
	}
}
System.out.println("Count of factors"+ c);
```

To check if a number (n) is Prime or not

```
int c=0;
for(int i=1; i<=n; i++){
	if(n%i==0){
	c++;
	}
}
if(c==2)System.out.println("Prime Number");
else System.out.println("Not Prime Number");
```

Factorial of a number (n), Written as n!
n!= 1 x 2 x 3 x 4 x 5 x …..x n

```
int f=1;
for(int i=1; i<=n; i++){
f=f*i;
}
System.out.println("Factorial = " +f);
```